

MNIST Maximum-Likelihood Classifier

Probability and Statistics — Final Project

1 Analytical Methodology

1.1 Class-Conditional Gaussian Model

Let $\mathbf{x} \in \mathbb{R}^d$ (with $d = 28 \times 28 = 784$) denote a flattened grayscale image and $y \in \{0, \dots, 9\}$ its digit label. The training set is $\mathcal{D} = \{(\mathbf{x}^{(n)}, y^{(n)})\}_{n=1}^N$ with $N = 60\,000$ examples. For each class c , let $\mathcal{D}_c = \{\mathbf{x}^{(n)} \mid y^{(n)} = c\}$ and $N_c = |\mathcal{D}_c|$.

We assume that images of digit c are drawn from a d -variate Gaussian:

$$p(\mathbf{x} \mid y = c) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}_c|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c)\right) \quad (1)$$

where $\boldsymbol{\mu}_c \in \mathbb{R}^d$ is the class mean and $\boldsymbol{\Sigma}_c \in \mathbb{R}^{d \times d}$ (symmetric, positive definite) is the class covariance matrix.

1.2 Maximum-Likelihood Parameter Estimation

For a fixed class c , the log-likelihood of the data in $\mathcal{D}_c$ is

$$\ell_c(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c) = -\frac{N_c}{2} \left[d \log(2\pi) + \log |\boldsymbol{\Sigma}_c| \right] - \frac{1}{2} \sum_{\mathbf{x}^{(n)} \in \mathcal{D}_c} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_c). \quad (2)$$

MLE for the mean. Setting $\nabla_{\boldsymbol{\mu}_c} \ell_c = \mathbf{0}$:

$$\hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{\mathbf{x}^{(n)} \in \mathcal{D}_c} \mathbf{x}^{(n)} \quad (3)$$

The MLE of the mean is the **sample mean** of all training images in class c .

MLE for the covariance. Differentiating (2) with respect to $\boldsymbol{\Sigma}_c^{-1}$ and equating to zero yields the sample (scatter) covariance:

$$\hat{\boldsymbol{\Sigma}}_c = \frac{1}{N_c} \sum_{\mathbf{x}^{(n)} \in \mathcal{D}_c} (\mathbf{x}^{(n)} - \hat{\boldsymbol{\mu}}_c)(\mathbf{x}^{(n)} - \hat{\boldsymbol{\mu}}_c)^\top \quad (4)$$

Since $d = 784$ is large, $\hat{\boldsymbol{\Sigma}}_c$ can be ill-conditioned. We regularise it with a small diagonal perturbation (Tikhonov regularisation):

$$\hat{\boldsymbol{\Sigma}}_c^{\text{reg}} = \hat{\boldsymbol{\Sigma}}_c + \varepsilon \mathbf{I}_d, \quad \varepsilon = 10^{-4}. \quad (5)$$

1.3 Prior Probabilities and MAP Decision Rule

Class priors are estimated by the empirical frequencies: $P(y = c) = N_c/N$, with $\sum_c P(y = c) = 1$.

Given a test image $\mathbf{x}$, we predict using the *maximum-a-posteriori* (MAP) rule. By Bayes' theorem and taking logarithms:

$$\begin{aligned}\hat{y} &= \arg \max_c [\log p(\mathbf{x} \mid y = c) + \log P(y = c)] \\ &= \arg \min_c \frac{1}{2} \log |\hat{\Sigma}_c| + \frac{1}{2} M_c(\mathbf{x}) - \log N_c\end{aligned}\quad (6)$$

where

$$M_c(\mathbf{x}) = (\mathbf{x} - \hat{\boldsymbol{\mu}}_c)^\top \hat{\Sigma}_c^{-1} (\mathbf{x} - \hat{\boldsymbol{\mu}}_c) \quad (7)$$

is the **squared Mahalanobis distance** between $\mathbf{x}$ and the class centroid $\hat{\boldsymbol{\mu}}_c$. The constant $\frac{d}{2} \log(2\pi)$ is dropped because it does not affect the $\arg \min$.

1.4 Implementation Pipeline

1. Normalise pixel intensities to $[0, 1]$.
2. For each class c : estimate $\hat{\boldsymbol{\mu}}_c$ (sample mean), $\hat{\Sigma}_c^{\text{reg}}$ (regularised sample covariance), and pre-compute $\hat{\Sigma}_c^{-1}$ and $\log |\hat{\Sigma}_c|$.
3. For each test image: compute $M_c(\mathbf{x})$ for all c and apply the MAP rule (6).

2 Experimental Results

The MNIST dataset contains 70 000 images (60 000 training, 10 000 test). Figure 1 shows example digits from the dataset, and Figure 2 shows the class distribution in the training set — all digits are roughly balanced ($\approx 10\%$ each).

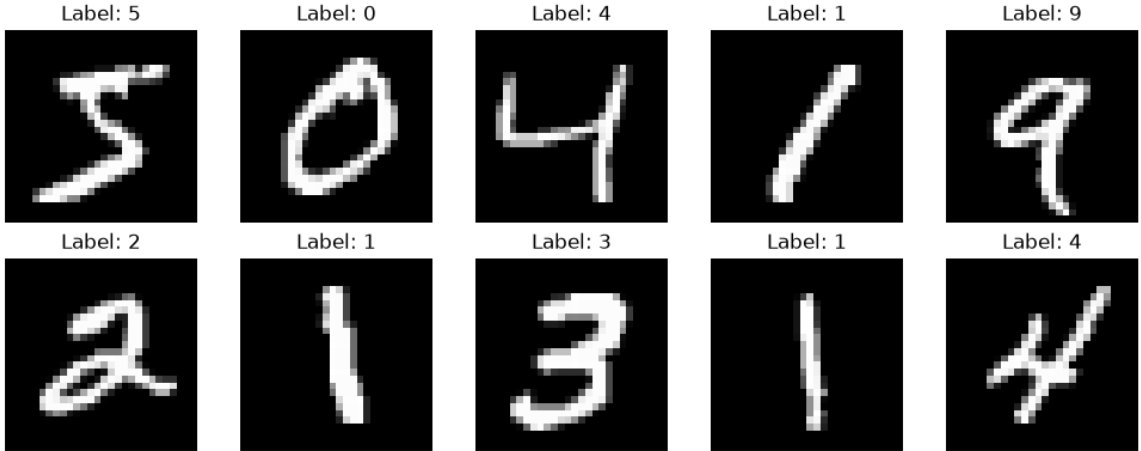


Figure 1: Example MNIST images with their labels.

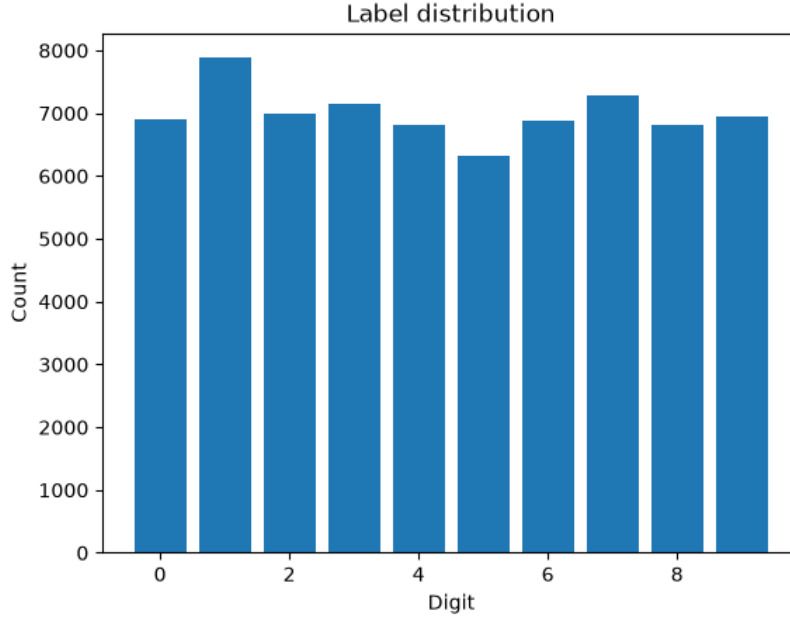


Figure 2: Label distribution in the training set (60 000 samples).

After normalising the pixel intensities to $[0, 1]$, we fit one Gaussian per digit class (10 Gaussians in $\mathbb{R}^{784}$) and classify the test set using the MAP rule (6).

2.1 Classification Performance

Table 1: Per-class and overall test accuracy.

Digit	0	1	2	3	4	5	6	7	8	9	All
Accuracy	95.3%	93.5%	87.6%	83.0%	81.7%	70.1%	93.9%	77.9%	88.6%	91.9%	86.6%
Test samples	986	1125	999	1020	975	902	982	1042	975	994	10 000

The overall test accuracy is **86.55%**. Digits 0, 6, and 1 are the easiest to recognise (above 93%), while digits 5 and 7 are the hardest (below 80%).

2.2 Confusion Matrix

Figure 3 shows the confusion matrix. The most frequent confusions are:

- Digit **5** $\rightarrow$ 3, 8 (many 5s are mistaken for 3s or 8s),
- Digit **7** $\rightarrow$ 2, 9 (7s often resemble 2s or 9s),
- Digit **4** $\rightarrow$ 9 (stylised 4s are confused with 9s).

These mistakes are consistent with human intuition: the digit pairs (5,3), (5,8), (7,2), (7,9), (4,9) share substantial visual similarity.

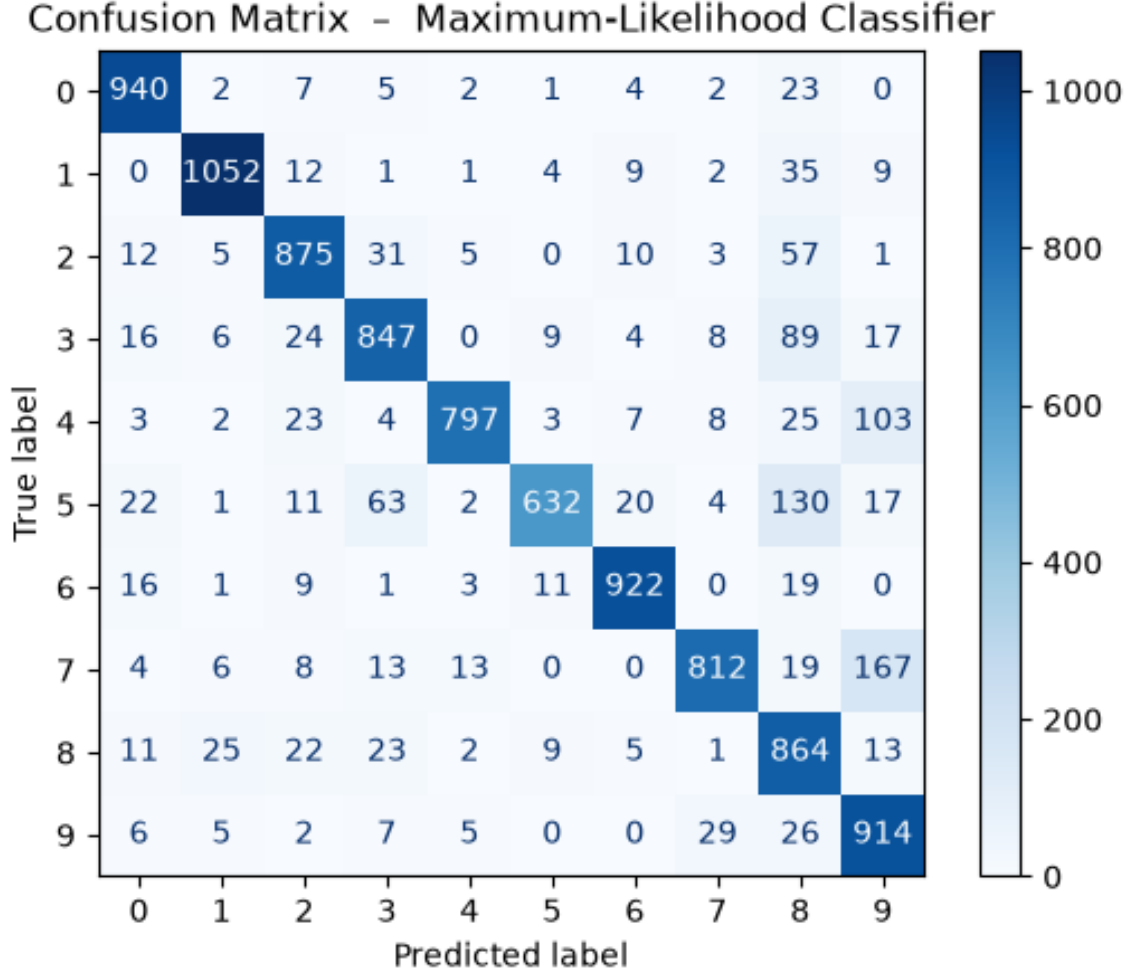


Figure 3: Confusion matrix of the maximum-likelihood classifier on the test set.

3 Discussion and Interpretation

Why does a simple Gaussian model work? Despite the high dimensionality ($d = 784$), a Gaussian per class is a surprisingly effective model. The class-conditional mean $\hat{\mu}_c$ captures the “prototypical” shape of each digit, while the covariance $\hat{\Sigma}_c$ captures the intra-class variability (stroke thickness, slant, translation). The MAP rule naturally balances how well the image matches the prototype (Mahalanobis distance) against the class prior.

Why does it fail on some digits? The Gaussian model assumes that the feature distribution inside each class is unimodal and ellipsoidal. Digits written in very different styles (e.g. “open” 4 vs. “closed” 4) form *multiple clusters* that a single Gaussian cannot capture. This explains the weaker performance on digits 5 and 7, which exhibit high stylistic variance. A Gaussian mixture model or a non-parametric density estimator would likely improve these digits.

Numerical considerations. The 784×784 covariance matrices require careful regularisation ($\varepsilon = 10^{-4}$) to remain invertible. Without regularisation, the matrices become numerically singular, especially for classes with fewer examples. The regularisation also acts as a weak Gaussian prior on the covariance (MAP estimation with an inverse-Wishart prior), which improves generalisation.

Comparison to baselines. An 86.6% accuracy is respectable for a purely probabilistic model with no feature engineering, no dimension reduction, and no deep learning. State-of-the-art convolutional networks achieve $>99\%$ on MNIST, but at the cost of millions of parameters and specialised architectures. The Gaussian classifier serves as an interpretable, statistically principled baseline.